

Semi-Passive RIS-Aided Super-Resolution DOA Estimation via Power-Domain Covariance Inversion and Difference Coarray Fusion

Chenchen Niu[†], Ye Tian[†], He Xu^{*}, Wei Liu[◇] and Tuo Wu[‡]

[†]*School of Artificial Intelligence, Ningbo University, Ningbo, China*

^{*}*School of Cyber Science and Engineering, Ningbo University of Technology, Ningbo, China*

[◇]*Department of Electrical and Electronic Engineering, Hong Kong Polytechnic University, Hong Kong*

[‡]*Department of Electronic Engineering, City University of Hong Kong, Hong Kong*

Abstract—In this paper, a super-resolution direction-of-arrival (DOA) estimation method in the non-line-of-sight (NLOS) scenario with a single receiving antenna is proposed, where a semi-passive reconfigurable intelligent surface (RIS) is deployed. A small number of low-cost auxiliary sensing antennas are embedded in the sparse RIS array, thereby enabling efficient and high-accuracy DOA estimation. Specifically, a beamforming-based approach that uses the output from the auxiliary sensing antennas is introduced for possible DOA range detection. Then, a high-dimensional spatial covariance matrix (SCM) is reconstructed by properly adjusting the RIS phase shifts and applying a Tikhonov-regularized least-squares inversion. Finally, difference-coarray lag fusion is exploited to enlarge the virtual array aperture and subsequently achieve super-resolution DOA estimation. Simulation results are provided to demonstrate the effectiveness and superiority of the proposed method.

Index Terms—DOA estimation, semi-passive RIS, single receiving antenna, power-domain covariance inversion, difference coarray fusion.

I. INTRODUCTION

Sensor array based direction-of-arrival (DOA) estimation plays an essential role in a wide range of applications, including localization, sensing, sonar, and wireless communications [1], [2]. Extensive research efforts have been devoted to DOA estimation under various application scenarios and hardware constraints, and many methods have been proposed, such as the subspace-based methods [3], [4], sparse reconstruction based methods [5], and deep learning based methods [6]. Nevertheless, most of these methods are developed under the assumptions of line-of-sight (LOS) or direct-path dominance, where phase-coherent multi-sensor observations are available and the signal model is well-conditioned.

However, with the rapid development of millimeter-wave (mmWave) and terahertz (THz) systems, the non-line-of-sight (NLOS) propagation becomes increasingly prevalent due to severe blockage and high path loss [7], [8]. In such scenarios, reliable DOA estimation is more challenging because the direct path may be absent or highly attenuated, and the remaining reflected or scattered components could be weak and unstable. Recently, reconfigurable intelligent surface (RIS) has been recognized as a promising means to create controllable virtual propagation paths, which can improve wireless access and

enable location-aware applications [9]. By adjusting the phase shifts of the impinging signals, RIS can reflect signals in a controlled way, thereby extending coverage and providing a possible way for DOA estimation in the non-line-of-sight (NLOS) scenarios [10].

Motivated by these advantages, a variety of RIS-aided DOA estimation methods have emerged in recent years. Beyond conventional beam scanning, advanced sparse and optimization-based approaches have been introduced, such as atomic norm minimization (ANM) [11] and semi-definite programming (SDP) [12] solutions, as well as the related covariance fitting [13] and low-rank recovery [14] techniques. While these methods demonstrate improved estimation performance, most existing designs still rely on either multi-antenna receivers or access to complex baseband snapshots after RIS combining, which may not be suitable for low-cost platforms. On the other hand, when the receiver is constrained to a single antenna and can only provide power-only measurements, the phase information is discarded and DOA estimation becomes more challenging.

To tackle this challenge, a semi-passive RIS-assisted super-resolution DOA estimation framework with a single receiving antenna is proposed in this paper. Different from traditional RIS structure, the adopted semi-passive RIS is composed of two parts: the first part is a small number of low-cost auxiliary sensing antennas, which is used to provide a coarse DOA range detection via the beamforming based approach; while the second part is a sparse RIS array, which is employed to reconstruct a high-dimensional spatial covariance matrix (SCM) at the single receiving antenna side by properly adjusting its phase shifts. Analysis and simulations show that the proposed method can provide an increased DOA resolution performance with low complexity.

II. SYSTEM MODEL

We consider an NLOS scenario where the direct link/path between the transmitter and the sensing receiver is blocked, and sensing is enabled by a semi-passive RIS, as shown in Fig. 1. The RIS follows an inverted-nested hybrid design and contains M reflecting elements in total. Specifically, it is composed of a sparse subarray with M_s elements spaced by $(M_d + 1)d$ and a dense subarray with M_d elements spaced by d , where $M = M_s + M_d$ and $d = \lambda/2$ denotes the half-wavelength spacing. Meanwhile, M_a low-cost auxiliary

This work was partially supported by the Ningbo Natural Science Foundation of China under Grant 62571286, the Zhejiang Provincial Ningbo Natural Science Foundation of China under Grant LQN26F010011, and the Ningbo Natural Science Foundation under Grant 2024J232

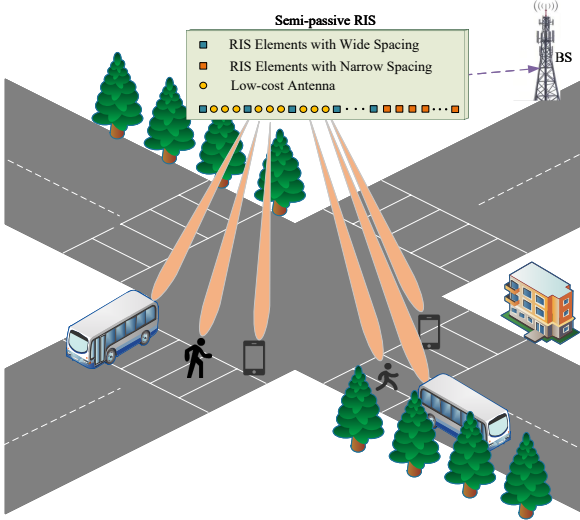


Fig. 1. Semi-passive RIS structure and its application in DOA estimation with a single receiving antenna.

sensing antennas are embedded in the inter-element gaps of the sparse subarray and are activated only for coarse DOA range detection. In the sensing mode, a single-antenna receiver acquires measurements under multiple RIS phase shifts.

Consider K narrowband far-field source signals impinging on the semi-passive RIS from DOAs $\theta = [\theta_1, \dots, \theta_K]^T$. Let $\mathbf{y}(\ell) \in \mathbb{C}^{M \times 1}$ represent the RIS output at time index ℓ , then it can be written as

$$\mathbf{y}(\ell) = \mathbf{A}_{\text{RIS}}(\theta) \mathbf{s}(\ell) + \mathbf{n}_r(\ell), \quad \ell = 1, \dots, L \quad (1)$$

where $\mathbf{A}_{\text{RIS}}(\theta) = [\mathbf{a}_{\text{RIS}}(\theta_1), \dots, \mathbf{a}_{\text{RIS}}(\theta_K)]$, and $\mathbf{a}_{\text{RIS}}(\theta) = \exp(j \frac{2\pi}{\lambda} \mathbf{p}_{\text{RIS}} \sin \theta) \in \mathbb{C}^{M \times 1}$ is the RIS array steering matrix with $\mathbf{p}_{\text{RIS}} = [p_1, \dots, p_M]^T \in \mathbb{R}^{M \times 1}$ denoting the positions of the RIS elements. $\mathbf{s}(\ell) \in \mathbb{C}^{K \times 1}$ and $\mathbf{n}_r(\ell) \in \mathbb{C}^{M \times 1}$ are the received source signals and additive Gaussian white noise, respectively.

The covariance matrix corresponding to $\mathbf{y}(\ell)$ is defined as $\mathbf{R} \triangleq \mathbb{E}\{\mathbf{y}(\ell)\mathbf{y}^H(\ell)\} \in \mathbb{C}^{M \times M}$, whose sampled valued with L snapshots is calculated as

$$\hat{\mathbf{R}} = \frac{1}{L} \sum_{\ell=1}^L \mathbf{y}(\ell) \mathbf{y}^H(\ell). \quad (2)$$

Similarly, let $\mathbf{p}_{\text{aux}} \in \mathbb{R}^{M_{\text{aux}} \times 1}$ stand for the positions of the M_{aux} auxiliary sensing antennas. As a result, the auxiliary array output and its matrix form can be written as

$$\mathbf{x}(\ell) = \mathbf{A}_{\text{aux}}(\theta) \mathbf{s}(\ell) + \mathbf{n}_a(\ell), \quad \ell = 1, \dots, L \quad (3)$$

$$\mathbf{X} = [\mathbf{x}(1), \dots, \mathbf{x}(L)] = \mathbf{A}_{\text{aux}}(\theta) \mathbf{S} + \mathbf{N}_a \quad (4)$$

where $\mathbf{A}_{\text{aux}}(\theta) = [\mathbf{a}_{\text{aux}}(\theta_1), \dots, \mathbf{a}_{\text{aux}}(\theta_K)]$ with $\mathbf{a}_{\text{aux}}(\theta) = \exp(j \frac{2\pi}{\lambda} \mathbf{p}_{\text{aux}} \sin \theta)$, $\mathbf{S} = [\mathbf{s}(1), \mathbf{s}(2), \dots, \mathbf{s}(L)]$ and $\mathbf{N}_a = [\mathbf{n}_a(1), \dots, \mathbf{n}_a(L)]$.

Let $\phi_q \in \mathbb{C}^{M \times 1}$ and $\mathbf{h}_r \in \mathbb{C}^{M \times 1}$ denote the RIS amplitude-phase shifting vector and the RIS-to-receiver channel vector,

respectively. Consequently, the output of the single receiving antenna can be expressed as

$$\begin{aligned} z_q(\ell) &= \mathbf{h}_r^T \text{diag}(\phi_q) \mathbf{y}(\ell) + v_q(\ell) \\ &= (\mathbf{h}_r \odot \phi_q)^T \mathbf{y}(\ell) + v_q(\ell) = \mathbf{w}_q^H \mathbf{y}(\ell) + v_q(\ell) \end{aligned} \quad (5)$$

where $\mathbf{w}_q = (\mathbf{h}_r \odot \phi_q)^* \in \mathbb{C}^{M \times 1}$, $v_q(\ell) = \mathbf{w}_q^H \mathbf{n}_r(\ell)$, subscript q represents the q -th phase shifting of the RIS, $\odot$ the Hadamard product, and $\text{diag}(\cdot)$ the diagonalization operation. Here, ϕ_q is a predetermined RIS control vector specified by the RIS controller at the q -th probing slot, while $\mathbf{h}_r$ is assumed to be known via prior calibration or pilot-assisted estimation. Consequently, $\mathbf{w}_q$ is available for the subsequent covariance reconstruction.

III. PROPOSED METHOD

In this section, we show how to exploit the special structure of the semi-passive RIS to achieve super-resolution DOA estimation from the perspective of power-domain covariance inversion and difference coarray fusion.

A. Coarse DOA Range Detection

For the low-cost auxiliary sensing antennas, complex radio frequency (RF) and digital signal processing unit configuration may be prohibitive in practice. In view of this, the beamforming based approach is adopted here, which constructs the following power spectrum over the entire DOA range $\theta \in [-90^\circ, 90^\circ]$:

$$P_{\text{BF}}(\theta) = \left\| \frac{\mathbf{a}_{\text{aux}}^H(\theta)}{\|\mathbf{a}_{\text{aux}}(\theta)\|_2} \mathbf{X} \right\|_2^2 \quad (6)$$

where $\|\cdot\|_2$ denotes the ℓ_2 norm.

According to the characteristic of beamforming, if θ corresponds to the true DOA, $P_{\text{BF}}(\theta)$ will exhibit a peak, which provides a fundamental solution for coarse DOA estimation. However, the beamforming based solution is constrained by the Rayleigh limit [15]. Taking this point into consideration, we here only employ it to detect possible DOA range. Under this guideline, each detected peak is expanded into a candidate DOA sector, given by

$$\mathcal{C}_m = [\hat{\theta}_m - \Delta, \hat{\theta}_m + \Delta] \quad (7)$$

where Δ controls the sector width. Overlapping sectors are merged to form the final candidate set $\mathcal{C} = \bigcup_{m=1}^{M_c} \mathcal{C}_m$.

B. Improved DOA Estimation

For the single receiving antenna, the sample-averaged received power can be calculated as

$$p_q = \frac{1}{L} \sum_{\ell=1}^L |z_q(\ell)|^2, \quad q = 1, \dots, Q. \quad (8)$$

According to the expression of $z_q(\ell)$ in (5), we have

$$p_q = \mathbf{w}_q^H \mathbf{R} \mathbf{w}_q, \quad q = 1, \dots, Q \quad (9)$$

Relying on $\mathbf{w}^H \mathbf{R} \mathbf{w} = (\mathbf{w} \otimes \mathbf{w}^*)^T \text{vec}(\mathbf{R})$, p_q can be written as

$$p_q = (\mathbf{w}_q \otimes \mathbf{w}_q^*)^T \mathbf{r} + \varepsilon_q, \quad (10)$$

where $\mathbf{r} = \text{vec}(\mathbf{R})$, ε_q is the transformed noise, and $\otimes$ represents the Kronecker product.

By adjusting RIS phase shifts to yield a series of p_q within the candidate set $\mathcal{C} = \cup_{m=1}^{M_c} \mathcal{C}_m$, we can construct a linear overdetermined system as

$$\mathbf{p} = \mathbf{\Psi} \mathbf{r} + \boldsymbol{\varepsilon}, \quad (11)$$

where $\mathbf{p} = [p_1, \dots, p_Q]^T \in \mathbb{R}^{Q \times 1}$, and the q -th row of $\mathbf{\Psi} \in \mathbb{C}^{Q \times M^2}$ is given by $(\mathbf{w}_q \otimes \mathbf{w}_q^*)^T$.

With available $\mathbf{p}$, the following Tikhonov-regularized least-squares optimization problem is formulated to reconstruct the high-dimensional SCM:

$$\hat{\mathbf{r}} = \arg \min_{\mathbf{r}} \|\mathbf{\Psi} \mathbf{r} - \mathbf{p}\|_2^2 + \delta^2 \|\mathbf{r}\|_2^2 \quad (12)$$

whose closed form solution is

$$\hat{\mathbf{r}} = (\mathbf{\Psi}^H \mathbf{\Psi} + \delta^2 \mathbf{I})^{-1} \mathbf{\Psi}^H \mathbf{p} \quad (13)$$

where δ^2 is a regularization parameter.

Notice that $\mathbf{R}$ holds a Hermitian structure, and therefore we can obtain the estimated $\hat{\mathbf{R}}$ by

$$\hat{\mathbf{R}} = \frac{\text{unvec}(\hat{\mathbf{r}}) + \text{unvec}(\hat{\mathbf{r}})^H}{2} \quad (14)$$

where $\text{unvec}(\cdot)$ denotes the inverse vectorization.

To enhance the effective aperture, we exploit the difference coarray induced by the RIS element locations $\{\xi_m\}_{m=1}^M$. Let $\hat{R}_{m,n}$ represent the (m, n) -th element of $\hat{\mathbf{R}}$, and $\Delta_{m,n}$ the normalized lag (in units of d), given by

$$\Delta_{m,n} \triangleq \xi_m - \xi_n, \quad m, n = 1, \dots, M. \quad (15)$$

Subsequently, all entries of $\hat{\mathbf{R}}$ corresponding to the same lag Δ are averaged to form the virtual correlation at lag Δ :

$$\hat{r}_v(\Delta) \triangleq \frac{1}{|\mathcal{I}(\Delta)|} \sum_{(m,n) \in \mathcal{I}(\Delta)} \hat{R}_{m,n} \quad (16)$$

where $\mathcal{I}(\Delta) \triangleq \{(m, n) : \xi_m - \xi_n = \Delta\}$ denotes the index set of all RIS element pairs, whose location difference equals to the lag Δ , and $|\mathcal{I}(\Delta)|$ is the number of covariance entries sharing the same lag, and (16) averages them to obtain the fused virtual correlation $\hat{r}_v(\Delta)$.

Let $\mathcal{D} \triangleq \{\Delta_{m,n}\}$ denote the resulting difference set. From $\{\hat{r}_v(\Delta)\}_{\Delta \in \mathcal{D}}$, we can extract the longest contiguous lag segment $\mathcal{U} = \{-U, -U+1, \dots, U\}$ and subsequently construct a virtual uniform linear array (ULA) covariance matrix as

$$\hat{\mathbf{R}}_v = \text{toep}([\hat{r}_v(0), \hat{r}_v(1), \dots, \hat{r}_v(U)]^T) \quad (17)$$

where $\text{toep}(\cdot)$ returns a Hermitian Toeplitz matrix utilizing the vector in the bracket.

With $\hat{\mathbf{R}}_v$, an eigen-decomposition is performed to yield

$$\hat{\mathbf{R}}_v = \mathbf{E}_s \mathbf{\Lambda}_s \mathbf{E}_s^H + \mathbf{E}_n \mathbf{\Lambda}_n \mathbf{E}_n^H \quad (18)$$

where $\mathbf{E}_s \in \mathbb{C}^{(U+1) \times K}$ and $\mathbf{E}_n \in \mathbb{C}^{(U+1) \times (U+1-K)}$ denote the signal and noise subspaces, respectively.

Finally, with the virtual-ULA steering vector

$$\mathbf{a}_v(\theta) = [1, e^{j \frac{2\pi}{\lambda} d \sin \theta}, \dots, e^{j \frac{2\pi}{\lambda} U d \sin \theta}]^T \quad (19)$$

improved DOA estimation is achieved by the following one-dimensional multiple signal classification (MUSIC) spectral function:

$$f(\theta) = \frac{1}{\mathbf{a}_v^H(\theta) \mathbf{E}_n \mathbf{E}_n^H \mathbf{a}_v(\theta)}. \quad (20)$$

It is emphasized that the construction of $\hat{\mathbf{R}}$ and the one-dimensional spectral search are all restricted to the small candidate DOA sector $\mathcal{C}$, and therefore, the proposed solution is computationally efficient.

IV. SIMULATION RESULTS

In this section, we examine the performance of the proposed method. through computer simulations, the RIS is configured with sixteen elements under a hybrid layout, and eight sparse elements spaced by $9d$ and eight dense elements spaced by d were deployed. Meanwhile, five low-cost auxiliary sensing antennas are embedded in the sparse inter-element gaps. The output spatial spectrums, probability of successful resolution (PoSR) for two closely spaced sources, and root mean square error (RMSE) of DOA estimation are employed to evaluate the performance.

In the first simulation, we show the output spectrums of the two-stage estimation for eight sources with their DOAs $\{-30.2^\circ, -25.4^\circ, -22.1^\circ, -18.7^\circ, 20.0^\circ, 23.5^\circ, 27.9^\circ, 31.5^\circ\}$. Q , L and SNR are set to 50, 500 and 0 dB, respectively. The first-stage beamforming based estimation result is shown in Fig. 2. It can be seen that the first-stage estimation yields two dominant broad peaks, which correctly indicates two angular clusters. Nevertheless, due to the limited aperture and the restriction of Rayleigh limit, the closely spaced sources within each cluster remain unresolved. Based on such two strongest peaks, we construct two candidate sectors with a half-width of $\Delta = 12^\circ$ for subsequent refinement; the corresponding output spatial spectrum is illustrated in Fig. 3, from which we can observe that the second-stage estimation can resolve all sources successfully, mainly due to the application of high-dimensional SCM reconstruction and difference-coarray lag fusion. In particular, the resulting spectrum exhibits eight sharp peaks that match the ground-truth DOAs, demonstrating high-resolution refinement with a significantly reduced search range and complexity.

In the second simulation, we further demonstrate the DOA resolution of the proposed method under different SNRs. Two closely spaced sources with their DOAs $\theta_1 = 24.5^\circ$, $\theta_2 = 25.5^\circ$ are considered, $Q = 20$ and $L = 500$. By assumption, two sources are successfully resolved if the estimation bias of two DOAs is less than 0.5° . For comparison, the physical-covariance $\hat{\mathbf{R}}$ based MUSIC estimator (designated here as PHY-MUSIC) is also considered. The simulation result is shown in Fig. 4. It can be seen that the proposed method

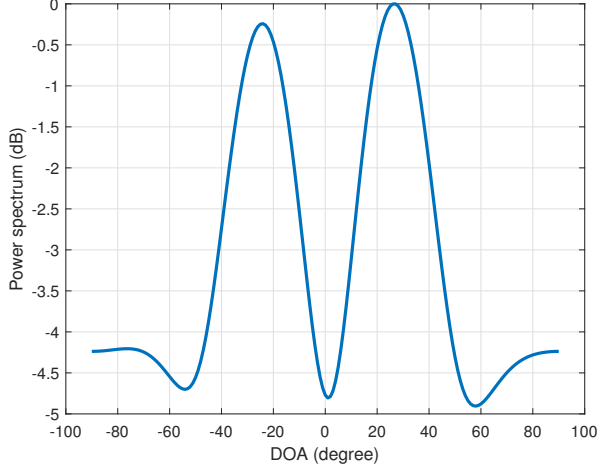


Fig. 2. Power spectrum obtained by the first-stage estimation, with $Q = 50$, $l = 500$ and $\text{SNR}=0$ dB.

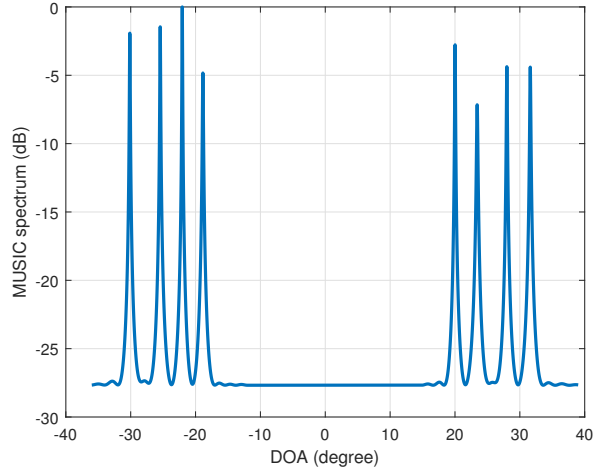


Fig. 3. Spatial spectrum obtained by the second-stage estimation, with $Q = 50$, $l = 500$ and $\text{SNR}=0$ dB.

reaches 100% correct resolution at a much lower SNR than the compared PHY-MUSIC method, again confirming its improved resolution performance.

In the last simulation, we assess the RMSE performance of the proposed method under different SNRs. The configurations of impinging DOAs are the same as in the first simulation, Q and L are respectively fixed at 50 and 500, respectively, while SNR is varied from -10 dB to 5 dB. As can be observed from Fig. 5, the RMSEs of both methods decrease with the increase of SNR. Interestingly, the proposed solution achieves a lower RMSE consistently and exhibits a clear accuracy gain in the medium-to-high SNR regime, demonstrating the benefit of the adopted virtual-aperture covariance enhancement strategy.

V. CONCLUSION

In this paper, a super-resolution DOA estimation method has been proposed for the NLOS scenario, jointly exploiting

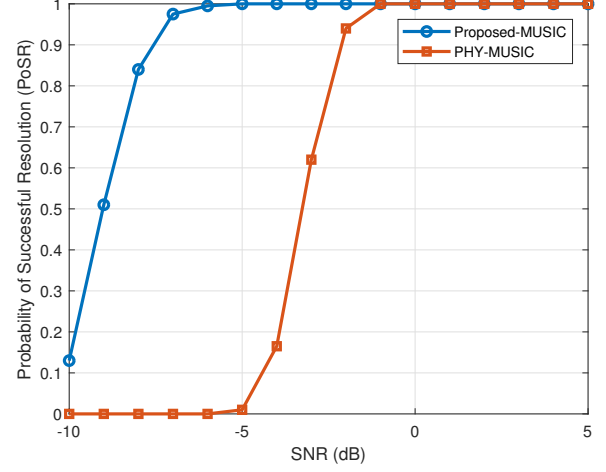


Fig. 4. PoSR versus SNR for two closely spaced sources, with $Q = 20$ and $L = 500$.

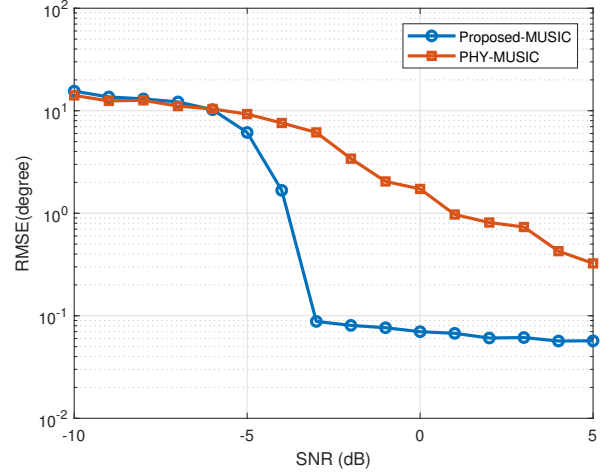


Fig. 5. RMSE versus SNR for eight sources, with $Q = 50$ and $L = 500$.

a semi-passive RIS and a single receiving antenna. By fully utilizing this special structure of the designed semi-passive RIS, a novel two-stage DOA estimation strategy is introduced to achieve efficient and low-cost estimation results, where the first-stage estimation is to obtain coarse DOA range detection by applying the beamforming based approach, while the second-stage estimation is to obtain improved and satisfactory DOA estimation result via high-dimensional SCM reconstruction and difference-coarray fusion. Simulation results demonstrate that the proposed solution can provide significantly improved DOA estimation resolution and accuracy.

REFERENCES

- [1] Y. Tian, X. Gao, W. Liu, H. Chen, G. Wang, and Y. Qin, "Localization of mixed far-field and near-field incoherently distributed sources using two-stage RARE estimator," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 2, pp. 1482-1494, Apr. 2023.

- [2] T. Tüker and O. Bialer, "Direction of arrival estimation and phase-correction for noncoherent subarrays: a convex optimization approach," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 6, pp. 5571-5585, Dec. 2022.
- [3] Z. Zheng, Y. Huang, W. -Q. Wang, and H. C. So, "Augmented covariance matrix reconstruction for DOA estimation using difference coarray," *IEEE Trans. Signal Process.*, vol. 69, pp. 5345-5358, 2021.
- [4] Y. Tian et al., "2-D DOA estimation of incoherently distributed sources considering gain-phase perturbations in massive MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 21, no. 2, pp. 1143-1155, Feb. 2022.
- [5] J. Dai and H. C. So, "Real-valued sparse Bayesian learning for DOA estimation with arbitrary linear arrays," *IEEE Trans. Signal Process.*, vol. 69, pp. 4977-4990, 2021.
- [6] Y. Tian et al., "Vehicle positioning with deep-learning-based direction-of-arrival estimation of incoherently distributed sources," *IEEE Internet of Things J.*, vol. 9, no. 20, pp. 20083-20095, Oct. 2022.
- [7] J. Li et al., "Mobility support for millimeter wave communications: opportunities and challenges," *IEEE Commun. Surv. Tutorials*, vol. 24, no. 3, pp. 1816-1842, Third Quart. 2022.
- [8] W. Jiang et al., "Terahertz communications and sensing for 6G and beyond: a comprehensive review," *IEEE Commun. Surv. Tutorials*, vol. 26, no. 4, pp. 2326-2381, Fourth Quart. 2024.
- [9] Z. Wang, Z. Liu, Y. Shen, A. Conti and M. Z. Win, "Holographic localization with synthetic reconfigurable intelligent surfaces," *IEEE J. Sel. Top. Signal Process.*, vol. 18, no. 4, pp. 603-618, May 2024.
- [10] Y. Tian, Y. Feng, W. Liu, H. Chen and G. Wang, "Reconfigurable intelligent surface aided DOA estimation by a single receiving antenna," *IEEE Trans. Commun.*, vol. 73, no. 2, pp. 983-994, Feb. 2025.
- [11] R. Li, S. Sun and M. Tao, "Atomic norm minimization-based DOA estimation for IRS-assisted sensing systems," *IEEE Wireless Commun. Lett.*, vol. 13, no. 10, pp. 2672-2676, Oct. 2024.
- [12] Y. Zheng, Q. Wang, L. Ren, Z. Ma and P. Fan, "RIS aided gridless 2D-DOA estimation via decoupled atomic norm minimization," *IEEE Trans. Veh. Technol.*, vol. 73, no. 10, pp. 15733-15738, Oct. 2024.
- [13] X. Wu, Z. Yang, Z. Wei, R. Schober and Z. Xu, "COFFEE: Covariance fitting and focusing for wideband direction-of-arrival estimation," *IEEE Trans. Signal Process.*, vol. 72, pp. 5659-5674, 2024.
- [14] A. Ertug Zorkun, M. A. Salas-Natera, R. Martínez Rodríguez-Orsorio and S. Chatzinotas, "Energy efficient low-complexity RIS-aided 3-D DOA estimation and target tracking algorithm via matrix completion," *IEEE Access*, vol. 12, pp. 197929-197941, 2024.
- [15] S. Li, Y. Liu, L. You, W. Wang, H. Duan, and X. Li, "Covariance matrix reconstruction for DOA estimation in hybrid massive MIMO systems," *IEEE Wireless Commun. Lett.*, vol. 9, no. 8, pp. 1196-1200, Aug. 2020.